

Task 6

Question:

Essay question – Propose how you will choose to design a conversational speech-to-speech pipeline in an essay no more than 700 words. Include what are some possible challenges you may face with your chosen design, and what are some ways to address them. Also include any references or materials you use for this task, which will not be part of the 700-word count. Your answer can be saved as essay-s2s.pdf under the main repository.

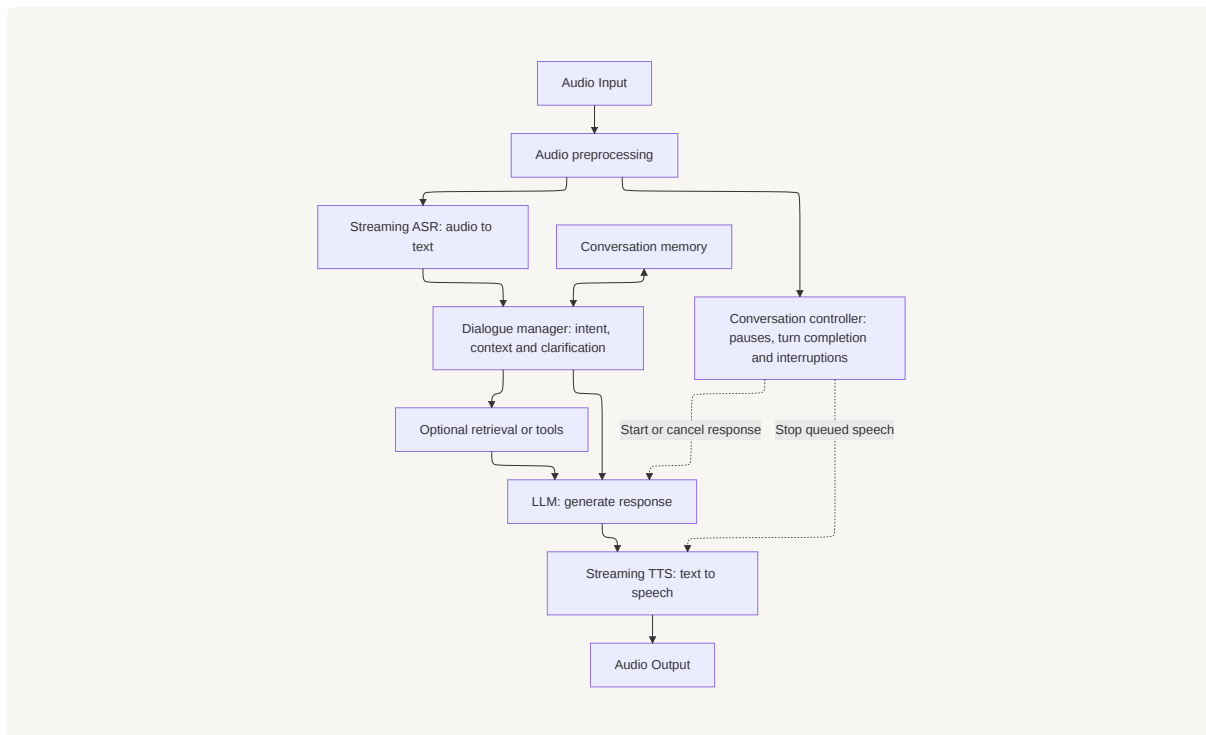
Designing a Conversational Speech-to-Speech Pipeline

I propose a streaming pipeline combining automatic speech recognition (ASR), a large language model (LLM), and text-to-speech (TTS), supported by a dialogue manager and conversation controller. Its modular design makes components easier to improve, replace and debug because errors can be traced to specific stages. Streaming reduces waiting by allowing information to be processed incrementally.

The pipeline contains four main components.

- Audio preprocessing would receive the speech signal, convert it into the required audio format, and reduce noise or echo.
- A streaming ASR would convert the speech into a transcript.
- The dialogue manager and LLM would identify the user's intent, maintain relevant conversation context, ask for clarification when necessary, and generate a response. An optional retrieval module could provide information from approved external sources.
- A streaming TTS would convert the response into speech.

A conversation controller would coordinate these components. It would detect turn completion, manage pauses, and handle interruptions. Importantly, streaming does not automatically make a system full-duplex. The system must continue listening while speaking and must be able to stop or replace its response when new information is received. The controller would therefore cancel queued speech and update the dialogue context when the user interrupts or corrects something.



For example, a caller to a proposed NUS admissions assistant might ask, "My daughter doing IB overseas, can apply or not ah?" ASR must recognise the informal phrasing and qualification. The assistant would clarify missing details, such as undergraduate or postgraduate study, the intended programme and admissions year, before retrieving relevant NUS guidance. It would explain the requirements without guaranteeing admission, pronouncing programme names, qualifications and dates clearly.

If the caller interrupts with, "Actually, she takes A-levels," the controller would cancel outdated generation and queued speech, update the context and process the correction. This illustrates why streaming alone does not ensure full-duplex interaction: the system must continue listening while speaking and respond to new information. Combining voice activity detection with a turn-completion model would help distinguish pauses and incomplete sentences from finished turns. Acknowledgements such as "mm-hm" should also be distinguished from interruptions.

Accents, informal speech and code-switching could cause transcription errors. I would test ASR using representative speech and request clarification when recognition is uncertain. For noisy or distant speech, I would test noise reduction and ask callers to repeat themselves when necessary. Echo cancellation, supported by separate input and playback audio streams, would reduce recognition of the assistant's own voice. Evaluation would compare

transcription accuracy and preservation of meaning across clean and noisy conditions, including false recognition during playback.

Retrieval introduces another risk: information may be irrelevant, outdated, missing or conflicting. I would restrict sources to trusted NUS pages, keep them updated, and filter results by programme, qualification and admissions year. When evidence is insufficient, the assistant should acknowledge uncertainty, clarify the question or refer the caller to an official contact. Source relevance, factual correctness and unsupported-answer rate would indicate whether these measures work.

To avoid long or unnatural replies, I would instruct the LLM to produce concise spoken responses and use phrase-level TTS. Keeping models loaded, streaming intermediate results, overlapping processing stages and limiting queue sizes would help reduce delays. I would assess answer length, pronunciation, intelligibility and naturalness, alongside median and 95th-percentile response latency. Relevant details, corrections and information already presented would be retained to maintain continuity across turns. Personal data collection would be minimised, access restricted and information deleted when no longer needed, with privacy and retention practices reviewed.

References

Reference	What it supports
<u>Moshi: a speech-text foundation model</u>	Native speech-to-speech as an alternative architecture
<u>DuplexCascade</u>	Full-duplex interaction using a coordinated streaming cascade
<u>Pipecat: Smart Turn</u>	Distinguishing speech pauses from turn completion
<u>Retrieval-Augmented Generation</u>	Supplying external information to the response generator
<u>NUS Office of Admissions</u>	Official information for the illustrative application
<u>Full-Duplex-Bench</u>	Evaluating interruptions, pauses, and turn-taking
<u>MTR-DuplexBench</u>	Evaluating multi-turn conversational behaviour